# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

# PHYSICS

Past Paper 2024 • Paper Code: 8471/8476

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Physics - Paper: II Objective (i)

طبیعیات

**Time Allowed:** 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8471

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Q1. One joule work done in moving a unit positive charge from one point to another in E is:

(A) 1 Watt
(B) 1 eV
(C) 1 Volt
(D) 1 Henry

Q2. The electric field intensity between two oppositely charged plates is:

(A) $\sigma/\varepsilon_0$
(B) $\sigma/\varepsilon_0$
(C) $\varepsilon_0/\sigma$
(D) $2\sigma/\varepsilon_0$

Q3. For Ohmic material, the current-voltage graph is:

(A) Parabolic
(B) Hyperbolic
(C) Straight
(D) Zigzag

Q4. Galvanometer is converted into voltmeter by connecting a high resistance "$R_h$" in:

(A) Series
(B) Parallel
(C) Bypass
(D) None of these

Q5. Force on a charge "q" in electric field E is:

(A) $F=q(v\times B)$ (B) $F=qB$

(C) $F=qE$ (D) $F=q(v\times E)$

Q6. The negative of potential gradient is called:

(A) Electric field intensity (B) Magnetic induction

(C) Electric potential (D) Electromotive force

Q7. In case of step up transformer:

(A) $Ns=Np$ (B) $Ns<Np$

(C) $Ns>Np$ (D) $Ns\neq Np$

Q8. The phase of negative peak of A.C. signal is:

(A) 0 rad (B) $\pi/2$ rad

(C) $2\pi$ rad (D) $3\pi/2$ rad

Q9. The maximum power loss in A.C. circuit is:

(A) $P=I_{rms}\,R$ (B) $P=I_{rms}\,V_{rms}$

(C) $P=I_{rms}\,V^2$ (D) $P=V_{rms}\,R$

Q10. The ability in a body to return to its original shape is called:

(A) Strain (B) Stress

(C) Plasticity (D) Elasticity

Q11. The current due to minority charge carriers in semiconductors is known as:

(A) Leakage current (B) Electronic current

(C) Conventional current (D) None of these

Q12. The size of base in a transistor is of the order of:

(A) $10^{-2}$ m (B) $10^{-3}$ m

(C) $10^{-4}$ m (D) $10^{-5}$ m

Q13. The value of speed of light is:

(A) $3\times10^{8}$ m/s (B) $3\times10^{-8}$ m/s

(C) $2\times10^{8}$ m/s (D) $2\times10^{-8}$ m/s

Q14. The Heisenberg uncertainty principle was proposed by W. Heisenberg in:

(A) 1928 (B) 1929

(C) 1927 (D) 1925

Q15. Helium was initially identified by using spectroscopy in:

(A) Moon (B) Sun

(C) Earth (D) Star

Q16. Dead time for Geiger-Muller counter is:

(A) $\sim 10^{-2}$ s (B) $\sim 10^{-3}$ s

(C) $\sim 10^{-4}$ s (D) $\sim 10^{-5}$ s

Q17. Energy released per nucleon during fission reaction is:

(A) 2 MeV (B) 3 MeV

(C) 1 MeV (D) 1.5 MeV

## Physics - Paper: II Subjective

## طبیعیات

**Time Allowed:** 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Q2.. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Electric lines of force never cross. Why?

(ii) Is E necessarily zero inside a charged rubber balloon, if balloon is spherical? Assume that charge is distributed uniformly over the surface.

(iii) Show that the region inside a hollow charged sphere is field free.

(iv) An electron has speed of $10^8$ m/s. Find its energy in electron volts.

(v) If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in the region is zero?

(vi) How can you use a magnetic field to separate isotopes of chemical element?

(vii) What is Lorentz force?

(viii) How can you make an electronic trajectory visible?

(ix) What do you understand by background radiation? State two sources of radiation.

(x) A particle which produces more ionization is less penetrating, why?

(xi) Define nuclear activity. Write its SI unit.

(xii) For what purpose Alcohol or Bromine is mixed with principle gas in Geiger tube?

Q3.. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Is the filament resistance lower or higher in a 500W, 220V light bulb than in a 100W, 220V bulb?

(ii) Describe a circuit which will give a continuously varying potential.

(iii) Define tolerance of a resistance and give an example.

(iv) What is the main reason for the world wide use of A.C?

(v) What is meant by A.M. and F.M.?

(vi) In an R-L circuit, will the current lag or lead the voltage? Illustrate your answer by a vector diagram.

(vii) Distinguish between crystalline, amorphous and polymeric solids.

(viii) Define modulus of elasticity. Show that the units of modulus of elasticity and stress are same. Discuss its three kinds also.

(ix) What is difference between ductile and brittle substances?

(x) What is meant by potential barrier? Give value of potential barrier for germanium and silicon.

(xi) Write down the symbol and truth table of exclusive NOR-gate.

(xii) What are the biasing requirements of the junctions of a transistor for its normal operation? Explain how these requirements are met in a common emitter amplifier?

Q4.. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) In a certain region the earth's magnetic field points vertically down. When a plane flies due north, which wingtip is positively charged?

(ii) When an electric motor, such as an electric drill, is being used, does it also act as a generator? If so what is the consequence of it?

(iii) As a solid is heated and begins to glow, why does it first appear red?

(iv) What happens to total radiation of a blackbody if its absolute temperature is doubled?

(v) What are advantages of Lasers over ordinary light?

(vi) On what factors mutual inductance depend?

(vii) What is Compton Shift? Give its mathematical expression.

(viii) At what angle Compton Shift is equal to Compton Wavelength?

(ix) Differentiate between ionization potential and excitation potential.

> Note: SECTION-C: (Attempt any THREE questions. Each question carries EIGHT (8) Marks)

Q5..

(a) [5] Define electric potential. Derive the relation of electric potential at a point due to a point charge.

(b) [3] $1.0\times10^{7}$ electrons pass through a conductor in 1.0 μs. Find the current in amperes flowing through the conductor. Charge on an electron is $1.6\times10^{-19}$ C.

Q6..

(a) [5] Derive the relation of torque on a current carrying coil.

(b) [3] A loop of wire is placed in a uniform magnetic field that is perpendicular to the plane of the loop. The strength of magnetic field is 0.6T. The area of the loop begins to shrink at a constant rate of $\Delta A/\Delta t = 0.8\ m^2s^{-1}$. What is the magnitude of emf induced in the loop while it is shrinking?

Q7..

(a) [5] Derive an expression for gain of an inverting OP-Amp and write down characteristics of OP-amplifier.

(b) [3] An iron core coil of 2.0 H and 50 Ω is placed in series with a resistance of 450 Ω. An A.C. supply of 100V, 50 Hz is connected across the circuit. Find (i) the current flowing in the coil, (ii) phase angle between the current and voltage.

Q8..

(a) [5] How does Davisson and Germer experiment confirm the wave nature of particles? Discuss in detail.

(b) [3] What stress would cause a wire to increase in length by 0.01% if the Young's modulus of the wire is $12\times10^{10}$ Pa? What force would produce this stress if the diameter of the wire is 0.56 mm?

Q9..

(a) [5] What are isotopes? How would you separate isotopes and find their masses using a mass spectrograph?

(b) [3] Calculate the longest wavelength of radiation for the Paschen Series.

## GROUP – II

## Physics - Paper: II Objective (iii)

طبیعیات

| **Time Allowed:** 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8476 |
|---|---|---|

> Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Q1. The phase difference between any two live phases of an AC supply is (Three phase supply):

(A) 120°
(B) 240°
(C) 120° or 240°
(D) 360°

Q2. At high frequency, voltage induced is maximum across:

(A) Resistor
(B) Capacitor
(C) Inductor
(D) RC Circuit

Q3. The use of commutator in D.C. motor may create ------ each time current reverses:

(A) Reverse torque
(B) Jerks
(C) Smooth running of armature
(D) Steady torque

Q4. If range of ammeter is to be increased, shunt resistance needs to be:

(A) Decreased
(B) Increased
(C) Unchanged
(D) Changed randomly

Q5. Self induction is used in:

(A) Capacitors
(B) Inductors
(C) Transformers
(D) Printers

Q6. The overall resistance of a voltmeter is:

(A) $R_h$R_g[illegible in source scan]
(B) $R_h$
(C) $R_h$+R_g
(D) $R_h$R_g/($R_h$+R_g)

Q7. Slope of current-voltage graph of Ohmic device gives (voltage is on x-axis):

(A) Resistance
(B) Conductance
(C) Power
(D) Energy

Q8. Gauss's law is applicable to:

(A) Closed path
(B) Open path
(C) Closed surface
(D) Flat surface

Q9. Equilibrium between electric and gravitational force is working principle of:

(A) Coulomb's law
(B) Gauss's law
(C) Millikan's method
(D) Faraday's law

Q10. The output of detector in mass spectrograph indicates:

(A) Number of isotopes
(B) Mass of isotopes
(C) Abundance of isotopes
(D) Accelerating potential

Q11. The difference in temperature of core of nuclear reactor and steam coming out of turbine is:

(A) 500°C
(B) 300°C
(C) 200°C
(D) 100°C

Q12. Inner shell transitions in a heavy metal lead to:

(A) Continuous X-rays
(B) Discrete X-rays
(C) Band X-rays
(D) Hard X-rays

Q13. The maximum value of Compton shift may be:

(A) 4.86 pm
(B) 2.43 pm
(C) 1.21 pm
(D) 9.68 pm

Q14. Best optical resolution achievable is:

(A) 0.2 nm
(B) 200 nm
(C) [illegible in source scan]
(D) 0.001 µm

Q15. The width of depletion region increases in a diode when it is:

(A) Unbiased
(B) Forward biased
(C) Reverse biased
(D) Normal biased

Q16. Commonly used semiconductor material is:

(A) Germanium
(B) Gallium Arsenide
(C) Silicon
(D) Selenium

Q17. Semiconductor behaves as perfect insulator in its pure form at:

(A) Room temperature
(B) 0 K
(C) High temperature
(D) Never

## Physics - Paper: II Subjective

## طبیعیات

**Time Allowed:** 2:40 Hours | **Maximum Marks:** 68

> Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Q2.. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Does electron tend to go to the region of high potential or of low potential?

(ii) Is E necessarily zero inside a charged rubber balloon if balloon is spherical? Assume that charge is uniformly distributed over the surface.

(iii) What is electric polarization? Does E decrease or increase due to polarization of a dielectric material?

(iv) What is capacitance of a capacitor? Define its unit.

(v) State Ampere's circuital law both in descriptive and mathematical form.

(vi) Describe the rule to find the direction of force on a current carrying conductor in magnetic field B.

(vii) Is it possible to orient a current loop in a uniform magnetic field such that the loop will not tend to rotate? Explain.

(viii) Why the resistance of an ammeter should be very low?

(ix) What factors make a fusion reaction difficult to achieve?

(x) What fraction of a radioactive sample decays after two half-lives have elapsed?

(xi) What do you mean by the term "Critical mass"?

(xii) If $^{233}_{92}U$ decays twice by α-emission, what is the resulting isotope?

Q3.. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Why is the potentiometer preferred over voltmeter? Explain.

(ii) A potential difference is applied across the ends of a copper wire. What is the effect on the drift velocity of free electrons by increasing the potential difference?

(iii) Is the filament resistance lower or higher in a 500W, 220V light bulb than in a 100W, 220V bulb?

(iv) What is a choke? Why is it used in A.C. circuits?

(v) A circuit contains an iron cored inductor, a switch and a D.C. source arranged in series. The switch is closed and after an interval reopened. Explain why a spark jumps across the switch contacts?

(vi) How does doubling the frequency affect the reactance of (a) an inductor (b) a capacitor?

(vii) At 0 K, a piece of Ge or Si is a perfect insulator. Why? Explain.

(viii) What do you mean by Curie temperature?

(ix) Distinguish between intrinsic and extrinsic semiconductors.

(x) Why is the base thin as compared to emitter and collector? Explain

(xi) Why do ordinary silicon diodes not emit light?

(xii) What is the principle of virtual ground? Apply it to find the gain of an inverting amplifier.

Q4.. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) When an electric motor, such as an electric drill, is being used, does it also act as a generator? If so, what is the consequence of this?

(ii) Describe the major causes of power loss in the transformer itself and suggest the remedies for these causes.

(iii) What is an electromagnet?

(iv) Will higher frequency light eject greater number of electrons than low frequency light?

(v) Which has the lower energy quanta? Radiowaves or X-Rays.

(vi) Calculate the minimum energy required for the pair production.

(vii) The cut off voltage for a certain metal is 100V. Find the work function for the cathode plate.

(viii) What do we mean when we say that the atom is excited?

(ix) What is holography?

> Note: SECTION-C: (Attempt any THREE questions. Each question carries EIGHT (8) Marks)

Q5..

(a) [5] Describe briefly how does Kirchhoff's explain the voltage rule.

(b) [3] A particle having a charge of 20 electrons on it, falls through a potential difference of 100 Volts. Calculate the energy acquired by it in electron volts.

Q6..

(a) [5] Discuss the principle, construction and working of a galvanometer.

(b) [3] Two coils are placed side by side. An emf of 0.8 V is observed in one coil when the current is changing at the rate of 200 A $s^{-1}$ in the other coil. What is the mutual inductance of the coils?

Q7..

(a) [5] Find the gain of a common emitter amplifier by using a transistor as amplifier.

(b) [3] Find the value of current flowing through a capacitance 0.5 μF when connected to a source of 150V at 50 Hz.

Q8..

(a) [5] Define intrinsic and extrinsic semiconductors. Explain p-type and n-type semiconductor materials also.

(b) [3] What is the de-Broglie wavelength of an electron whose kinetic energy is 120 eV?

Q9..

(a) [5] Write down a note on working, uses and construction of Geiger-Muller counter.

(b) [3] An electron jumps from a level $E_p=-3.5\times10^{-19}$ J to $E_r=-1.20\times10^{-18}$ J. What is the wavelength of emitted light?